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***B.Tech. Degree I Semester Regular and Supplementary/
II Semester Supplementary Examination November 2018***

**EC GE 15 – 1103 B ENGINEERING GRAPHICS
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

(5 × 12 = 60)

- I. A length of 600 m is represented by a line of 30 cm in a drawing. Draw a backward reading Vernier scale to read upto 400 m. Draw the scale for a least count of 50 cm. Mark on the scale the length of 205.5 m and 166.5 m.

OR

- II. The major axis of an ellipse is 150 mm long and the minor axis is 100 mm long. Find the foci and draw the ellipse by arc of circle or intersecting arc method. Draw a tangent and normal to the ellipse at a point on it.
- III. The front view of a line AB makes an angle of 30° with the X-Y line, whose VT is 30 mm above HP. The end A is 10 mm below HP and end B is 100 mm behind VP. Draw the projections of the line if HT is 40 mm above HP. What is the true length and true inclination of the line?

OR

- IV. Draw projections of a pentagonal lamina of 30 mm side having one of its sides in the HP and inclined at 65° to the VP and its surface making an angle of 50° with the HP.
- V. A square pyramid side of base 45 mm and altitude 65 mm is kept with a side of base parallel to VP and the triangular face containing that side of base being vertical. Draw the projections of the pyramid such that the base is visible in the front view.

OR

- VI. A hexagonal pyramid of base side 25 mm and axis 60 mm rests on its base on HP with two base edges perpendicular to VP. It is cut by a plane perpendicular to the VP and inclined at 30° to the HP, meeting the axis at 25 mm from the vertex. Draw sectional plan and true shape of the section.
- VII. A vertical square prism, base 50 mm side is completely penetrated by a horizontal square prism, base 35 mm side, so that their axes are 6 mm apart. The axis of the horizontal prism is parallel to the VP while the faces of both the prisms are equally inclined to the VP. Draw the projections of the prisms showing line of intersection.

OR

- VIII. A pentagonal prism edge of base 30 mm and height 60 mm is placed on HP so that one of its base edge parallel to VP. A horizontal hole is drilled on the solid so that the axis of the hole and axis of the prism bisect each other. The axis of the hole is perpendicular to VP of 36 mm diameter and having a Draw the development surface of the solid with the hole.

- IX. Draw the isometric view of a pentagonal pyramid side of base 40 mm and height 80 mm which rests centrally on a cylinder of diameter 120 mm and height 40 mm.

OR

- X. Draw the perspective projection of a pentagonal prism of side 25 mm and length 50 mm, lying on one of its rectangular faces on the ground plane and one pentagonal face touching the picture plane. The station point is 55 mm in front of the picture plane and lies in a centre plane which is 75 mm to the left of the centre of the prism. Station point is 30 mm above the ground plane.

